

ARTICLE

Understanding market sentiment analysis: A survey

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Abstract

Market sentiment analysis (MSA) has evolved significantly over nearly four decades, growing in relevance and application in economics and finance. This paper extensively reviews MSA, encompassing methodologies ranging from lexicon-based techniques to traditional Machine Learning (ML), Deep Learning (DL), and hybrid approaches. Emphasizing the transition from rudimentary word counters to sophisticated feature extraction from diverse sources such as news, social media, and share prices, the study presents an updated state-of-the-art review of sentiment analysis. Furthermore, using network analysis, a bibliometric and scientometric lens is applied to map the expanding footprint of sentiment research within economics and finance, revealing key trends, dominant research hubs, and potential areas for interdisciplinary collaboration. This exploration consolidates the foundational and emerging methods in MSA and underscores its dynamic interplay with global financial ecosystems and the imperative for future integrative research trajectories.

KEYWORDS

Deep Learning, Financial Sentiment Analysis, Machine Learning, Stocks

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JEL CLASSIFICATION

G02; G14; L86

1 | INTRODUCTION

Artificial Intelligence (AI), Big data, and social media are drastically changing our world, and the finance sector is no exception. One of the wonders of the stock market is that although markets may seem efficient, share prices can change regardless of the discounted value of the assets and the technologies a company holds. Market sentiment has become an important factor contributing to share prices and is highly influenced by the news and trends in social media. Thus, market cap, a measure of the value of a company in the market, can become higher or lower than the company's tangible assets.

Sentiment analysis and opinion mining are terms that are interchangeably used to refer to a field of study that concludes a product or organization can be affected by people's views, emotions, and attitudes. Such analysis and mining are increasingly extracted from content on the web. Little was done in the field prior to the turn of the millennium; however, sentiment analysis has now found profound uses in management sciences, political sciences, economics, and social sciences, as such areas are affected by people's opinions. It has achieved high accuracy rates in E-commerce product reviews and rose to become one of the major research areas in Natural Language Processing (NLP) (Liu & Zhang, 2012). It can be shown that market sentiment affects the price trend, trade volume, volatility, and potential risks (Azimi & Agrawal, 2021; Kazemian et al., 2016; Kearney & Liu, 2014; Tetlock, 2007).

A recent group decision on the options market by hundreds of thousands of investors on Reddit, a trendy social media group, increased the price of a stock share, GameStop, multiple times and caused billions of dollars of damage to traditional Hedge funds. The broker, Robinhood, was forced by the authorities to stop these investors trading, and as a result, the price of GameStop dropped. Corbet et al. (2022) show that such behaviors can be identified by a lexicon of terminologies of keywords used in explicit and implicit manipulation attempts. This will prevent the messaging board from being used to manipulate markets (Corbet et al., 2022).

The effect of news on the market price can be practically observed. For instance, before Joe Biden, the President of the United States, made a positive speech, announcing a \$2.25T investment in electric vehicles, renewable power and the electric grid today (March 31, 2021), market prices had already turned bullish on the S&P 500, following a month of more or less bearish trends on most stocks.¹ And a few posts (tweets) by Elon Musk of Tesla affected the company's share price and also the price of the cryptocurrency Bitcoin significantly.²

Forecasting Sentiments and share prices can be performed by applying AI to features extracted from news, social media including X (Twitter), Facebook, Instagram and Reddit, and the share price history. Sentiment analysis systems are usually coarse-grained and focus on explicit sentiment, which does not suit topic-specific or company-specific sentiments in news articles. MSA should ideally be able to analyze both explicit and implicit trends (Van de Kauter et al., 2015). Furthermore, a word might have different meanings in various contexts. Within this context, the following research presents an overview of the relatively brief history and the latest advances in market sentiment analysis and aims to address the financial terms in a simple form for computer scientists. Despite the advancements in market sentiment analysis methodologies, several persistent challenges affect the accuracy and reliability of specific rating systems. These include the inherent subjectivity in sentiment interpretation, potential biases in source data, and the dynamic

nature of market contexts, which may render static models less effective. Moreover, the linguistic nuances and contextual variations often lead to misclassifications, particularly in sarcasm and implicit expressions. Additionally, the rapid evolution of market terminologies and the emergence of unconventional data sources, such as social media, introduce complexities that traditional sentiment analysis models struggle to address. This section delves into these challenges, offering insights into the limitations of current rating systems and suggesting potential avenues for future research to mitigate these issues.

This investigation reveals the dominant role of select nations, particularly the United States, China, and the United Kingdom, in driving sentiment-related research in economics and finance. Through analysis, central themes such as market dynamics, investor behavior, and risk assessment have been underscored, with particular keywords including “market”, “investor”, “return”, “index”, and “risk” recurrently emerging. The research emphasises the adaptability of sentiment analysis in integrating current global events, as evidenced by the prominence of terms, for example, “COVID” in recent literature. Despite its evident significance, a conspicuous gap is noted in the representation of sentiment analysis within high-tier academic publications, suggesting potential discrepancies between traditional academic perceptions and the innovative essence of this research domain. This study serves as a comprehensive map of the prevailing landscape of sentiment-related studies in these disciplines and a testament to the potential of sentiment analysis in shaping future economic and financial inquiries.

The examination of sentiment-related research in economics and finance reveals key focal points: a noticeable divergence between theoretical constructs and practical market implications and an under-representation of sentiment analysis in top-tier academic journals. While nations such as the United States, China, and the United Kingdom prominently steer sentiment research, there is a risk of overlooking insights from emerging economies. The adaptability of sentiment analysis to global shifts underscores its potential as a primary analytical tool. These insights emphasize the need for broader academic collaboration, redefining scholarly validation standards, and harnessing sentiment analysis for nuanced financial understanding and foresight.

The rest of this paper proceeds as follows: Section 2 presents a brief history of expert systems in financial analysis, while Section 3 presents an overview of the data and the methodological approaches employed. Section 4 presents the empirical results, while Section 5 presents an associated discussion, implications and related directions for future research. Finally, Section 6 concludes.

2 | BRIEF HISTORY OF EXPERT SYSTEMS IN FINANCIAL ANALYSIS

Birts et al. (1993), reviewed the interest in expert systems for financial analysis and traced it back over a decade. They found that the finance sector had not widely accepted or used expert systems, as users had not perceived a need to use them. The team suggested a re-orientation in the development of expert systems. Bouwman (1983), identified the lack of access to supporting arguments for final decisions in an understandable and user-friendly manner to users as a major complaint about decision-making computers, specifically asking student analysts to verbalize their thoughts at each stage to humanize the decision-making process. These verbal traces, referred to as “protocols”, were then analyzed at various levels. The analysis of these protocols led to the development of models representing the decision-making process, implied strategies, and task-specific financial knowledge. These models and strategies were then converted into executable computer programs, and extensive comparisons and evaluations of the computational model were carried out. Despite

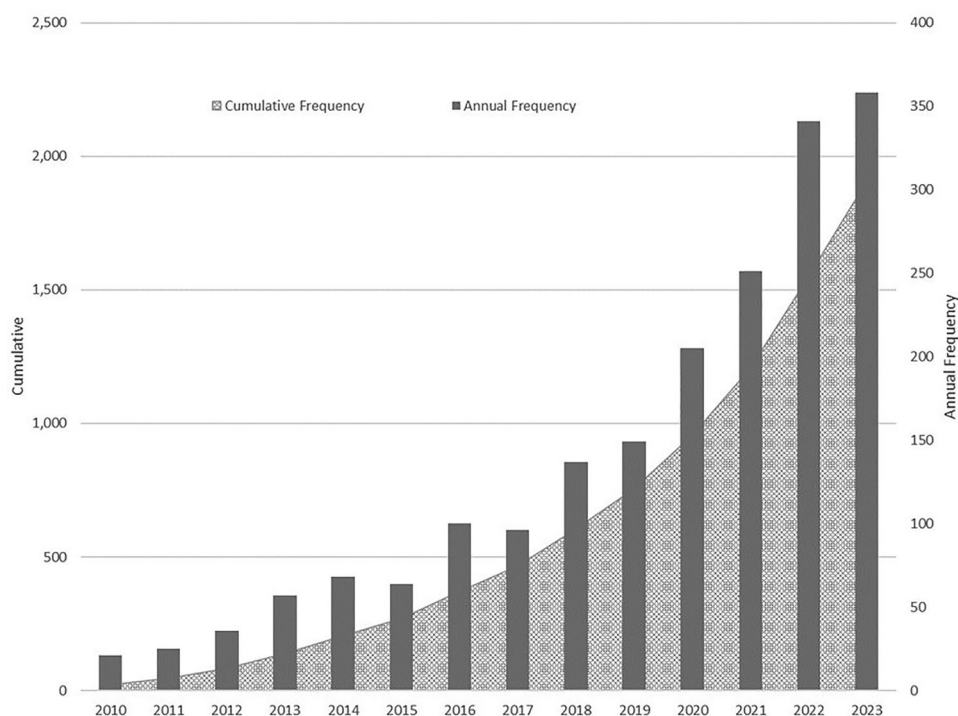


Figure 1 Frequency of Publication: Economics & Finance-based Research Based on Sentiment Analysis.

Note: All data were downloaded from Scopus on June 30, 2023 as both CSV and plaintext, where all information was selected. This allows for the analysis of inward and outward citations, abstracts and various other bibliometric areas. Core bibliometric approaches involve surfacing the linkages between papers or articles.

their algorithm lacking “linguistic rigidity”, the authors drew similar inferences and conclusions to those made by human decision-makers (Bouwman, 1983).

To understand the concepts of Sentiment Analysis (SA) and Natural Language Processing (NLP), we consider a few key components. Market sentiment, which can be bullish or bearish, is determined by the trends of increasing or decreasing prices. However, short-term market prices can also be influenced by factors such as emotions, uncertainties, or attempts to inflate stocks, that is, to pump stocks artificially. NLP has become very popular for fundamental analysis, as many macroeconomic factors are effectively unstructured, making it possible to capture sentiment more accurately, with further nuances of the semantics (Xing et al., 2018). In NLP, the “Bag-of-Words” technique is employed, where a collection of words from a sentence or a document provides a simplified representation of the text (Gondaliya et al., 2021). Fundamental analysis is a method used to measure a security’s intrinsic or fair value. It is based on basic economic and financial principles, macroeconomic values such as the sector’s condition, and microeconomic parameters, including the efficiency of the firm’s management (Man et al., 2019).

It is informative to understand the growth of research that has utilized SA in economics and finance in recent years. In Figure 1, we observe the annual frequency and cumulative frequency of research focusing on SA, explicitly published in the subject areas of economics and finance. Such research has grown exponentially in recent years, with value predominantly based on the interactions between sentiment and stock market performance (Akyildirim et al., 2020, 2021; Cioroianu et al., 2021a, 2021b, 2023; Corbet et al., 2020; Kyriazis et al., 2020, 2023).

3 | DATA AND METHODOLOGICAL APPROACH

MSA systems have historically been applied to sources including news, blogs, and social media data, including Twitter (now X) and Facebook posts. The advantage of news and social media is their high frequency, compared to, for example, low-frequency company annual reports although the latter may include a wider range of data which can be very informative (Man et al., 2019). The validity of the data depends on the source. The different news channels might report the same incident differently due to their business plan and possible biases, the knowledge of their team of journalists on the specific subject, the terms that they use, and their sensitivity to market trends. Thus, when reading text data sourced from news or social media, including the source and user profile would be helpful. Low-frequency corporate financial disclosures do not contain enough data for time series modeling due to the low frequency of the reports. However, the qualitative information they include may be very informative. Daudert et al. (2018) and Daudert and Buitelaar (2018) have combined the data from news and social media, where such data tends to be noisy and unregulated, but the information is frequent and has been used extensively in sentiment analysis (Akyildirim et al., 2020, 2020, 2021, 2021; Ranco et al., 2015; Rao et al., 2014; Smailović et al., 2013; Zhang et al., 2018).

Several papers have utilized various sources of data, including Wikiposit, which includes 16 sources: Stocks (NASDAQ, DJIA, S&P 500, Nikkei 225, Hang Seng index, FTSE100, DAX), ASX Currencies (EUR, AUD, JPY, USD) and Commodities (Silver, Platinum, Oil, Gold) (Shen et al., 2012).³

With further academic value, this research further provides a network analysis of the key research that has utilized sentiment analysis in economics and finance to understand best how such research has developed over time. We utilize the Scopus database as our source of records as this captures the widest range of papers with complete reference sets and both author and institution metadata in a consistent form.⁴ We chose 2010 as a starting point for the research, as further back in any bibliometric database⁵, the coverage becomes more sparse (Harzing & Alakangas, 2016; Michels & Schmoch, 2012). Finally, all citation measures, unless otherwise noted, include self-citation.⁶ As per Corbet et al. (2019), we further estimate the applicability of Lotka's Law (Chung & Cox, 1990) to the dataset. Lotka's law suggests that the number of publications by authors is best described as an inverse square law. Lotka's Law is formulated as $A = K/X^n$, where K and n are constants. Usually, $n = 2$ is the number of authors publishing n papers, and X represents the number publishing one paper. This implies that the number of authors publishing X is a fixed ratio of two to the number of authors publishing a single article. We used the R package Bibliometrix (Aria & Cuccurullo, 2017) for this analysis.⁷ Network analyses were carried out using <https://www.vosviewer.com/> VOSviewer, which is a software tool that is used to construct and visualize bibliometric networks.⁸ The search strategy⁹ used was:

```
(TITLE-ABS-KEY (sentiment analysis)
AND PUBYEAR > 2010
AND (LIMIT-TO (DOCTYPE,"ar"))
AND (LIMIT-TO (SUBJAREA,"ECON" OR "FIN"))
AND (EXCLUDE (PREFNAMEAUID, "UndefinedUndefined"))
```

(1)

All data were downloaded from Scopus on June 30, 2023 as both CSV and plaintext, from which all information was selected. This allows for the analysis of inward and outward citations,

abstracts, and other bibliometric areas. Core bibliometric approaches involve uncovering the linkages between papers or articles. Here, we use, unless otherwise indicated, the number of articles to weigh collaboration and linkage.¹⁰ Linkages are unless otherwise noted, based on the number of documents. The thicker the connecting lines in the networks, the higher the weight. The package VosViewer was used for this analysis, supplemented by Gephi¹¹ and the R package Bibliometrix¹² (Aria & Cuccurullo, 2017).

Following completion of the data collection process, we next separate the collated publications based on the Chartered Association of Business Schools (ABS), Australian Business Deans Council (ABDC), and SCImago Journal Rank (SJR) journal rankings. While multiple issues have been identified¹³, the use and analysis of each of these journal ranking systems present observations based on the average of these rankings. The <https://charteredabs.org/academic-journal-guide-2021/> ABS list has been stated “to assist researchers in making informed judgements about the outlets they may wish to publish in” with the ambition to provide detail on a wide range of journals across multiple fields related to business and management studies; in other words. Research quality is identified as 4* (highest) to 1 (lowest). The <https://abdc.edu.au/abdc-journal-quality-list/> ABDC list is ranked from A* (highest) to C (lowest). The ABDC’s mission is to “make Australian business schools better and to foster the national and global impact of Australian business education and research”. This is seen to involve being the collective and collegial voice of university business schools while providing opportunities for members to share knowledge and best practices. The <https://www.webofscience.com/wos/woscc/basic-search> Web of Science (WoS) is a paid-access platform providing access to multiple databases that provide reference and citation data from academic journals, conference proceedings, and other documents in various academic disciplines. It was originally produced by the Institute for Scientific Information. From WoS, the Journal Citation Reports (JCR) is an annual publication integrated with the Web of Science and is accessed from the Web of Science Core Collection. It provides information about academic journals in the natural sciences and social sciences, including impact factors. Finally, the <https://www.scimagojr.com/journalrank.php> SCImago Journal Rank (SJR) is a measure of the prestige of scholarly journals that accounts for both the number of citations received by a journal and the prestige of the journals where the citations come from. It is provided as a numeric value representing the average number of weighted citations received during a selected year per document published in that journal during the previous 3 years; therefore, the higher the SJR indicator values, the greater a journal’s estimated prestige.

4 | EMPIRICAL RESULTS

Sentiment analysis approaches can be broadly classified into dictionary or lexicon-based, which work based on word count, and Machine Learning methods. The latter can be further divided into traditional ML, such as Support Vector Machine (SVM), Random Forest (RF), and Deep Learning (DL), which learns a sequence of embeddings. Hybrid methods also exist (Araci, 2019; Cortis et al., 2017). Lexicon-based methods create a dictionary where each word represents a polarity of positive, negative, or neutral sentiments towards a subject. The meaning of some of the words depends on the context. There are also abbreviations and noisy text. Thus, high-order processing and intelligent analysis techniques are necessary, which can be addressed using ML (Ravi & Ravi, 2015). It has been shown that negative words are more powerful in explaining market variations (Tetlock et al., 2008). A term-weighting scheme can improve the importance of the positive words (Jegadeesh & Wu, 2013; Kearney & Liu, 2014).

ML methods use algorithms such as SVM, Recurrent Neural Network (RNN), Convolutional Neural Network (CNN), Random Forest, Naïve Bayes (NB), and Long Short-Term Memory (LSTM) (Kuko & Pourhomayoun, 2020). Hybrid models combine traditional ML and DL with lexicon-based feature extraction and word and sentence encoders. Currently, Natural Language Processing is extensively used (Mishev et al., 2020). There are different layers of the analysis: Document level, Sentence level, Entity and aspect level (several entities and their different aspects describe opinion targets) (Liu & Zhang, 2012).

Schumaker and Chen (2009) used a bag of words to find the textual keywords to determine the effect of news on market price using Genetic Algorithms, Naïve Bayes and Support Vector Regressors (SVR). They applied three ML algorithms to 9,211 financial news articles and 10,259,042 stock quotes covering the S&P 500 stocks during 5 weeks. The model containing article terms and the stock price at the time of article release had a similar price change direction as the future price (57.1%) (Schumaker & Chen, 2009). Bollen et al. (2011) extracted Twitter data using Google Profile of Mood States and Opinion Finder and achieved an accuracy of 87.6%. Shen et al. (2012) used SVM to exploit the temporal correlation between the global stock market and various financial products to get the next-day trend. They used the Wikiposit database SVM and Multiple Additive Regression Trees (MART) algorithms. They reported that feature selection was essential to improve SVM results, but MART required less training data and a high-dimensional feature set. The SVM results were 74.4%, 76% and 77.6% for the cases of NASDAQ, S&P500, and DJIA, respectively. Using the top four features, SVM and MART yield 74.4% and 70.3%; using the full features, they achieve 63.1% and 73.9% (Shen et al., 2012). Smailović et al. (2013) used SVM to classify Tweets into positive, negative, and neutral categories and showed that the posts expressing a public opinion on companies and their products can predict the stock price a few days earlier. In addition to Twitter-specific text preprocessing, they performed N-grams of size two. Lee et al. (2014) determined the importance of text analysis and showed that linguistic features (unigrams) were preferred over financial features, and the effect of sentiment diminishes with time. They mentioned that a security's price was highly dependent on corporate reports in the short term and relied on third-party opinion, reflected in newspapers in the long term, as more time was required for the market to digest newspaper articles. Table 1 shows the summarized methods and results obtained from these discussed research pieces.

Kalyanaraman et al. (2014) used a Bing API to extract word frequency in news articles matched against a dictionary of positive and negative categories implemented within a linear regression model, with gradient descent for optimization. The accuracy of the stock price model was higher than the accuracy of the sentiment. Cakra and Trisedya (2015) used a linear regression predictive model to classify Tweets in the Indonesian market into positive, negative, and neutral, disregarded neutral Tweets, and retrieved sentiment lexicons semi-automatically, where informal words were checked manually. They used a simple sentiment analysis model with different classification techniques, where the Random Forest algorithm produced better results over Naïve Bayes. Van de Kauter et al. (2015) provided a fine-grained method focused on topic-specific and company-specific content and detected both explicit and implicit sentiments. Pagolu et al. (2016) used N-gram and Word2vec (2-layer NN) to analyse sentiment polarity in Tweets and concluded that more data increases accuracy. Xu and Cohen (2018) presented the state-of-the-art method, a Neural Network (NN) architecture for price prediction based on Tweets and historical share prices. StockNet, a deep generative model, demonstrates and introduces recurrent, continuous latent variables to tackle the stochastic aspect of data. Gondaliya et al. (2021) applied six different ML algorithms, including , Decision Tree (DT), KNN, Logistic Regression (LR), Naïve Bayes, Random Forest, and SVM to the Indian stock market for market prediction covering the recent

Table 1 Comparison of the different methods.

Researcher(s)	Database	Feature(s)/ Method(s)	Result(s)
Schumaker and Chen (2009)	S&P 500, News	SVR	57.1%
Bollen et al. (2011)	DJIA, Twitter	Google Profile of Mood States and Opinion Finder	87.6%
Shen et al. (2012)	NASDAQ, S&P 500, DJIA	SVM	74.4%, 76%, 77.6%
Smailović et al. (2013)	DJIA, Twitter	SVM, Twitter-specific text preprocessing, and N-grams of size 2	81.06%
Lee et al. (2014)	Volatility index (VIX), S&P 500	Baseline1, Baseline2, Uni: Unigram model, NMF model, Ensemble model	55.5 % for Ensemble model
Kalyanaraman et al. (2014)	Bing API dataset	Linear Regression with Gradient Descent	Positive: 59.53%, Negative: 59.63
Cakra and Trisedya (2015)	Twitter, hybrid features	SVM, NB, DT, RF, NN to classify tweets for sentiment analysis. Linear Regression used for prediction	66.34%, 57.43%, 66.34%, 65.35%, 65.35% sentiment. Prediction: 0.9989 and 0.9983.
Pagolu et al. (2016)	Twitter	Logistic regression train, LibSVM	69.01%, 71.82%
Xu and Cohen (2018)	Twitter, and historical prices	Deep generative model	58.23%
Gondaliya et al. (2021)	Indian stock, and Twitter, News, RSS feed	DT, KNN, LR, NB, RF, SVM (Bag-of-words)	77%, 53%, 78%, 62%, 76%, 78%
Chou et al. (2021)	AAPL, Investors' sentiment	DL prediction model, LSTM recurrent NN	74%
Kaur et al. (2021)	IEEE data port/COVID Twitter data	DL, RNN, H-SVM, SVM	Positive, negative, neutral sentiments spotted
Štrimaitis et al. (2021)	Four Lithuanian business news websites	NB, LSTM, SVM	71.1%, 71%, 70.4%

Note: MSA systems have historically been based on data sources such as News, blogs, and social media data, including Twitter and Facebook posts, for example, and corporate disclosures. The advantage of News and social media is their high frequency, compared to company disclosures, while company disclosures, such as the company's annual reports, are low frequency but include more qualitative information and can be very informative (Man et al., 2019). The validity of the data depends on the source. The different news channels might report the same incidence differently due to their business plan and possible biases, the knowledge of their team of journalists on the specific subject, the terms that they use, and their sensitivity to market trends. Thus, while reading the text data from news or social media, it would be helpful to include the source and user profile, too.

outbreak of COVID-19. They reported that Logistic Regression and SVM showed superiority over the other algorithms where the bag-of-words technique was applied. Chou et al. (2021) proposed a deep-learning stock prediction approach based on investor sentiment to tackle the volatility and noise issues during the COVID-19 pandemic. They implemented an LSTM recurrent neural network for its ability to analyze time series through a memory function and utilized an atten-

Table 2 The most prevalent nations of affiliation of published works and the most commonly used keywords in research based on Sentiment Analysis.

Rank	Country	Docs	Cites	Link Strength	Keyword	Count	Relevance
1	United States	374	7,951	187	Market	466	0.7821
2	China	304	2,992	168	Investor	376	1.143
3	United Kingdom	181	2,668	166	Return	324	1.6827
4	India	150	959	37	Index	280	1.4298
5	Germany	117	1,009	68	Risk	232	0.5752
6	Australia	87	1,372	83	Stock Market	228	1.3767
7	Spain	82	1,053	50	Volatility	219	1.6736
8	Italy	79	439	42	Price	212	0.7125
9	Russia	70	332	26	News	189	0.5751
10	Taiwan	66	504	47	Social Medium	171	0.676
11	France	59	722	70	Review	165	1.3182
12	South Korea	54	1,096	14	Text	151	0.9386
13	Canada	45	633	41	System	149	0.4341
14	Turkey	45	544	35	COVID	148	0.2437
15	Greece	44	394	25	Consumer	137	1.0247
16	Malaysia	42	362	30	Crisis	121	0.5176
17	Hong Kong	38	819	45	Sentiment Index	112	1.5936
18	Poland	34	355	17	Shock	110	1.4306
19	Portugal	30	228	16	Uncertainty	105	0.8571
20	Netherlands	29	812	26	Algorithm	104	0.8793

Note: All data were downloaded from Scopus as both CSV and plaintext, where all information was selected. This allows for the analysis of inward and outward citations, abstracts, and other bibliometric areas.

tion function to model the critical information. They report that this model could predict the prices, regardless of the uncertainty, during the pandemic. Kaur et al. (2021) presented an algorithm in R language for analyzing Twitter hashtags, including COVID-19, coronavirus, deaths, new cases, and those recovered. They applied a deep learning strategy, RNN, SVM, and Hybrid Heterogeneous Support Vector Machine (H-SVM) to spot positive, negative and neutral sentiments. Štrimaitis et al. (2021) apply supervised ML to four news websites in Lithuania, considering the complexities of the Lithuanian language compared to English. Naïve Bayes, LSTM, and SVM algorithms yielded accuracy estimates of 71.1%, 71% and 70.4%, respectively.

Focusing on the prevalence of sentiment-related research in economics and finance subject domains, Table 2 presents the most prevalent nations and words associated with the affiliations and indexed keywords, respectively. Universities in the United States, China and the United Kingdom are identified to be the most interlinked in terms of relevance, to have been published focusing on the area of sentiment analysis, while the terms “market”, “investor”, “return”, “index”, and “risk” are the most commonly used keywords. Table 3 provides evidence of the prevalence and growth of such research annually between 2010 and 2023. Although the frequency of occurrence for such terms might be greater, such a list combines the relevance score when considering such ranking. It is interesting to note the prevalence of terms such as COVID, with reference to the recent global pandemic, which has become commonplace in economics and finance research utilized sentiment analysis despite its relatively short lifespan. The most frequently used keywords indicate that most research using such foci surrounding financial market effects concentrated on

Table 3 The most prevalent nations of affiliation of published works and the most commonly used keywords in research based on Sentiment Analysis Over Time.

Rank	Country	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
1	United States	7	5	11	18	17	18	26	25	29	31	32	41	54	60
2	China	3	0	4	2	5	3	9	15	9	21	31	47	64	94
3	United Kingdom	2	3	3	5	14	5	10	11	12	16	22	15	27	36
4	India	0	1	0	4	0	2	10	3	13	7	13	31	40	32
5	Germany	2	1	4	4	3	5	6	7	10	8	10	13	21	23
6	Australia	1	1	3	4	5	2	7	5	6	8	12	9	12	13
7	Spain	0	0	1	4	2	4	3	2	4	12	9	11	14	16
8	Italy	0	2	0	1	2	3	5	6	2	7	8	7	19	18
9	Russia	0	0	1	1	1	2	2	2	6	8	7	19	15	6
10	Taiwan	2	3	3	3	4	4	5	4	3	6	4	6	11	8
11	France	0	0	1	0	2	1	0	5	3	1	7	6	17	17
12	South Korea	1	1	1	0	4	2	0	1	3	4	11	8	9	9
13	Canada	0	0	1	1	0	0	0	2	6	6	11	9	6	3
14	Turkey	0	1	1	0	0	2	0	3	2	4	8	5	9	10
15	Greece	0	1	1	3	1	2	4	2	2	4	2	9	6	7
16	Malaysia	1	0	1	0	2	1	3	3	3	2	5	3	8	10
17	Hong Kong	0	1	2	2	1	2	1	3	0	3	5	5	8	9
18	Poland	0	1	1	2	2	1	0	1	4	4	5	6	4	3
19	Portugal	0	1	0	1	0	0	2	1	1	2	4	6	4	8
20	Netherlands	1	2	0	2	0	1	0	3	1	3	5	2	3	6
Rank	Keyword	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
1	Market	11	16	20	28	36	34	55	47	66	82	102	120	156	176
2	Investor	10	7	15	16	26	21	36	28	44	39	75	70	115	133
3	Return	6	3	12	10	24	20	32	21	32	33	46	54	83	82
4	Index	2	1	8	9	7	7	14	13	18	23	41	50	55	74
5	Risk	5	0	6	6	10	12	16	15	22	26	35	44	66	78
6	Stock Market	2	4	7	8	8	13	15	17	21	21	35	40	58	58
7	Volatility	4	2	10	5	9	9	11	17	13	8	24	39	44	55
8	Price	2	6	12	10	15	13	21	15	29	26	42	54	79	70
9	News	2	2	5	5	5	15	13	11	20	24	25	41	57	46
10	Social Medium	0	0	0	0	0	0	0	7	9	8	7	14	13	18
11	Review	11	16	20	28	36	34	55	47	66	82	102	120	156	176
12	Text	3	2	5	13	10	24	22	20	32	34	61	76	108	103
13	System	3	5	3	11	10	6	10	10	11	11	25	38	43	60
14	COVID	0	0	0	0	0	0	0	0	0	0	5	33	55	50
15	Consumer	2	4	5	6	10	8	20	11	18	27	30	30	49	40
16	Crisis	0	1	1	6	8	7	7	13	9	13	21	23	29	27
17	Sentiment Index	19	22	31	53	63	58	89	87	126	135	190	231	313	338
18	Shock	0	0	2	4	3	6	5	6	7	4	15	19	25	18
19	Uncertainty	0	0	3	4	4	1	3	5	6	8	11	19	31	31
20	Algorithm	0	1	1	3	3	3	2	6	5	7	14	16	28	20

Note: All data were downloaded from Scopus as both CSV and plaintext, where all information was selected. This allows for the analysis of inward and outward citations, abstracts, and other bibliometric areas.

price volatility and liquidity effects. Examining the data presented in Tables 2 and 3, several noteworthy patterns emerge that provide insights into the evolution and direction of sentiment-related research within the domains of economics and finance. To begin, the dominance of universities from the United States, China, and the United Kingdom in sentiment analysis research underscores the significance of these regions in driving academic discourse in this area. Several factors could account for this. First, the prominence of financial markets within these nations positions them as natural front-runners in exploring the complex interplay between sentiment, investment behaviors, and market outcomes. Furthermore, these nations house many of the world's leading financial hubs, where rapid information dissemination and market reactions underscore the value of sentiment analysis.

Focusing specifically on the indexed keywords reveals an inherent inclination towards examining market dynamics. The emphasis on terms such as “market”, “investor”, “return”, “index”, and “risk” suggests that the primary concern for scholars is understanding the behavioral aspects of market participants and the consequential market outcomes. These terms, taken together, signify a focus on the mechanics of the financial market, with a particular interest in gauging how investor sentiments translate into tangible market indicators, such as returns and risk. Interestingly, the advent of unforeseen global events, as evidenced by the mention of “COVID”, elucidates the adaptability and responsiveness of the academic community. Despite its recent emergence, the prominence of COVID-related keywords underscores the magnitude of its impact on global economic and financial landscapes. While the term's relative novelty might make its prominence surprising, the pandemic's unprecedented challenges to global economies and financial systems rendered it an indispensable research subject. The inclusion of COVID in sentiment analysis research indicates an attempt to understand the immediate and long-term repercussions of global shocks on market sentiments and their cascading effects on financial metrics. Further, the recurrence of research themes related to price volatility and liquidity effects corroborates the idea that sentiment analysis remains closely tethered to understanding market stability and fluidity. Given the critical importance of these metrics in shaping policy decisions and investment strategies, it is evident that the academic community perceives sentiment analysis as a pivotal tool in deciphering the intricacies of market dynamics.

In Figure 2, we observe a network analysis based on overall abstract co-occurrence. Sentiment analysis and effect remain separated as two research clusters, directly interlinked between terms based on the context and theoretical underpinnings of such work and both the econometric and market-focused effects of such work, respectively. The distinction between “sentiment analysis” and “effect” as two separate research clusters indicates a bifurcation in academic interest: one rooted in the conceptual and theoretical foundations of sentiment and the other delving into its tangible repercussions in econometrics and market dynamics. This separation can be attributed to the nuanced intricacies each cluster delves into, necessitating specialized methodologies and theoretical frameworks. Figure 3 reiterates the central position of the United States within the main research clusters, with direct interaction with both China and the United Kingdom. Such evidence suggests the presence of quite a centralized core of authors focusing on such works. Smaller nations appear relatively isolated from this core research cluster. The dominance of the United States as a central node, with pronounced connectivity to both China and the United Kingdom, underscores the collaborative nature of research endeavors among these academic powerhouses. Their centrality can be attributed to their pioneering roles in advancing financial research and housing major financial hubs. This central cluster, however, leaves lesser-researched nations on the periphery, highlighting potential areas where collaboration could be broadened to integrate diverse perspectives and enrich broad research. Keyword co-occurrence is presented in Figure 4, where investor sentiment and broad sentiment analysis appear separated in two differential clus-

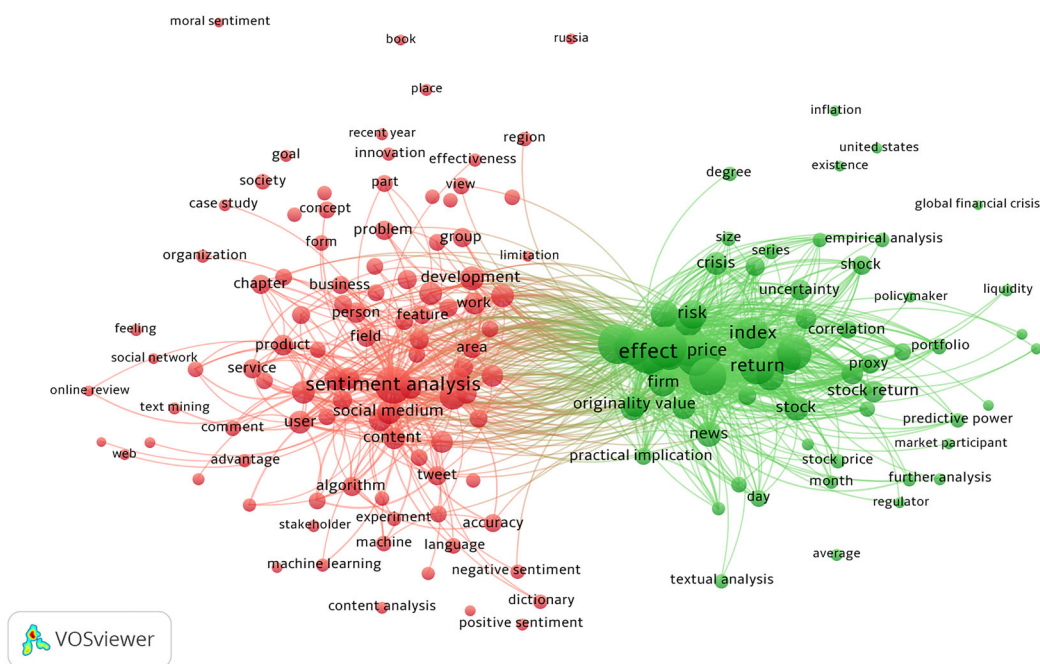


Figure 2 Overall abstract cooccurrence.

[Colour figure can be viewed at wileyonlinelibrary.com]

Note: The above network analysis was carried out using <https://www.vosviewer.com/VOSviewer>, which is a software tool that is used to construct and visualize bibliometric networks. These networks can be constructed based on citation, bibliographic coupling, co-citation, or co-authorship relations. VOSviewer also offers a text mining functionality that can be used to construct and visualize co-occurrence networks of important terms extracted from a body of scientific literature.

ters. COVID-19 presents a distinct central node, along with terms such as sustainability, finance, corporate social responsibility, consumer behavior, media, and risk assessment. The keyword co-occurrence patterns shed light on the prevailing concerns and intersections in the domain. The distinct clusters for “investor sentiment” and broader “sentiment analysis” underscore the specialized nature of investigations. The emergence of “COVID-19” as a focal node interlinked with themes such as “sustainability”, “corporate social responsibility”, and “consumer behavior” mirrors the pandemic’s profound influence, necessitating a comprehensive exploration across diverse research streams, ranging from corporate governance to risk evaluation.

We next separate data based on the journal ranking that has been associated with each publication in Table 4. Based on each ranking, it is evident that research based on sentiment analysis has not been commonplace in those academic outputs determined to be of the highest quality as per ABDC, SJR, and ABS. Otherwise, research appears to be quite evenly spread across each of the other quality-determined groups.¹⁴ Several factors could account for this observation. Predominantly, the rigorous editorial standards and highly selective nature of these premier journals often prioritise well-established, traditional research methodologies over emerging interdisciplinary approaches, such as sentiment analysis. This conservatism ensures the reliability and reproducibility of published findings but may sometimes lead to a lag in embracing nascent fields. Moreover, the interdisciplinary nature of sentiment analysis, intertwining finance, economics, psychology, and data analytics, might render it challenging to cater to the specific readership of these top-tier journals, which typically expect content grounded firmly within their specialized domain. The perception of sentiment analysis, despite its objective foundations in data analytics,

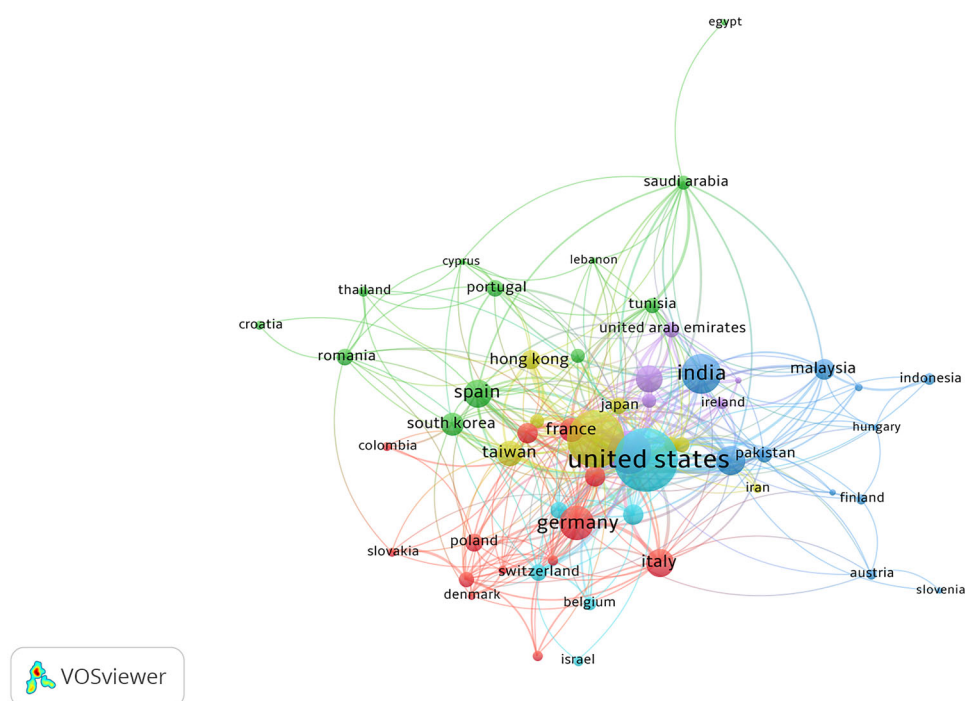


Figure 3 Geographical interconnectivity between published authors.

[Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/joes.12564)]

Note: The above network analysis was carried out using <https://www.vosviewer.com/VOSviewer>, which is a software tool that is used to construct and visualise bibliometric networks. These networks can be constructed based on citation, bibliographic coupling, co-citation, or co-authorship relations. VOSviewer also offers a text mining functionality that can be used to construct and visualize co-occurrence networks of important terms extracted from a body of scientific literature.

may be seen as relatively subjective, especially when juxtaposed against the conventional quantitative methods frequently celebrated in these journals. Furthermore, the even spread of sentiment analysis research across other quality tiers indicates a broader acceptance and recognition of its value within the academic community. It also underscores an opportunity for these methodologies to mature and refine further, eventually finding their rightful place in the echelons of top-tier publications in due course. As the field consolidates its empirical methodologies and offers more conclusive findings, it is plausible that its representation in elite journals will witness an uptick.

Affiliation location and associated keyword link strength are ranked as per the frequency of occurrence in each of the ABS, ABDC, and SJR ranking systems in Tables 5, 6, and 7, respectively. It is evident that United States-based affiliations consistently outperform other nations in high-ranking academic outputs in the topic areas of economics and finance throughout each of the considered ranking systems, adding further robustness to the data collection and separation structures. Further, a tiered structure is identified when considering the keywords that are listed, with those relating to stock market analyses consistently positioned at the top of each presented list irrespective of the ranking system considered. It is notable that the link strength, as estimated over the period 2010 through 2023, appears to incorrectly estimate the presence of recent works focusing on broad sectoral review and the development of sentiment indices. The observed tiered keyword structure, emphasizing stock market analyses, underscores the global influence of the US financial markets and their centrality in shaping global economic narratives. Such prominence not only underscores the weight given to US markets in academic literature but also reflects the

earnings management

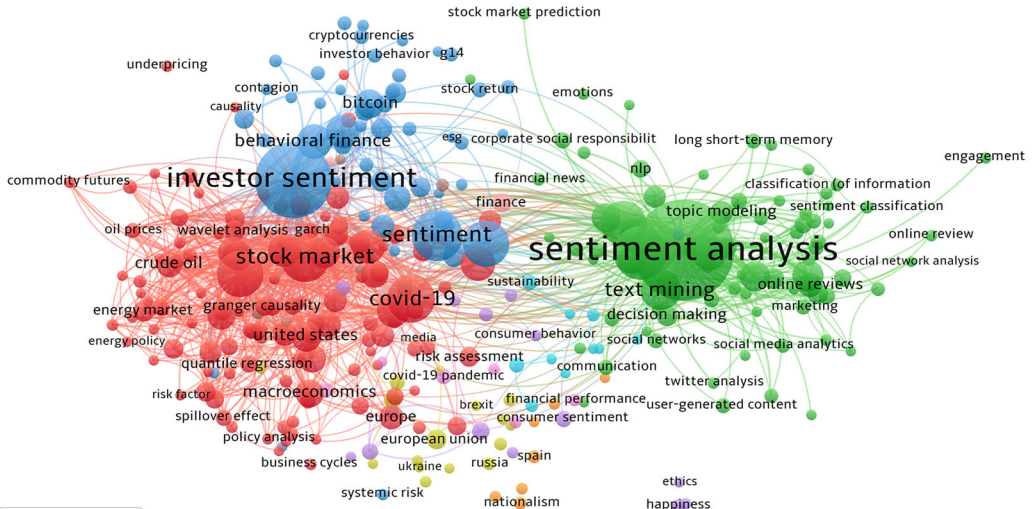


Figure 4 Overall keyword cooccurrence.

[Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/joes.12544)]

Note: The above network analysis was carried out using <https://www.vosviewer.com/VOSviewer>, which is a software tool that is used to construct and visualize bibliometric networks. These networks can be constructed based on citation, bibliographic coupling, co-citation, or co-authorship relations. VOSviewer also offers a text mining functionality that can be used to construct and visualize co-occurrence networks of important terms extracted from a body of scientific literature.

Table 4 Total journal ranking summary statistics (by year), unique sentiment analysis-based research in economics and finance journals.

	Rank	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	Total
ABDC	A*	0	1	2	5	2	2	7	3	5	4	10	7	17	68
	A	0	6	14	4	20	12	16	32	34	32	58	72	72	452
	B	9	9	21	12	27	48	21	36	36	33	48	42	78	579
	C	8	4	12	12	16	20	40	12	16	16	12	56	60	356
SJR	Q1	0	3	1	3	1	2	5	2	2	1	0	1	1	22
	Q2	8	2	14	10	30	24	20	14	20	20	24	10	8	204
	Q3	15	18	21	15	15	18	21	33	12	30	30	36	33	297
	Q4	0	8	20	12	20	20	4	16	20	20	28	20	16	204
ABS	4*	0	0	0	0	1	0	1	1	1	3	2	1	0	10
	4	0	0	0	2	1	0	2	0	2	3	0	0	0	10
	3	0	0	3	6	6	14	11	27	36	15	26	17	14	175
	2	0	0	3	15	15	21	12	17	10	9	12	13	22	149
	1	0	0	0	0	8	20	12	12	12	4	8	12	8	96

Note: Data presented represent the number of unique publications, with associated citations presented in parentheses. All data were downloaded from Scopus as both CSV and plaintext, where all information was selected. This allows for the analysis of inward and outward citations, abstracts, and other bibliometric areas. Core bibliometric approaches involve surfacing the linkages between papers or articles. This data is subsequently separated and collated based on each of the Chartered Association of Business Schools (ABS), Australian Business Deans Council (ABDC), Web of Science (WoS), and SCImago Journal Rank (SJR) journal rankings lists.

Table 5 Abstract and keyword occurrence, and geographical distribution of unique economics and finance publications (as ranked by ABS).

Rank	Country	4*	4	3	2	1	Keyword	4*	4	3	2	1
1	United States	9	6	25	33	24	Market	8	4	52	62	68
2	China	1	1	5	20	6	Investor	2	5	43	37	45
3	United Kingdom	3	2	17	10	11	Return	3	4	38	32	30
4	India	0	0	1	3	11	Index	1	0	17	12	16
5	Germany	1	0	7	6	10	Risk	2	3	17	12	20
6	Australia	1	1	4	5	10	Stock Market	1	0	12	22	24
7	Spain	0	0	6	4	1	Volatility	1	1	19	15	15
8	Italy	1	0	3	6	3	Price	1	2	18	23	35
9	Russia	0	0	0	1	0	News	1	3	14	10	12
10	Taiwan	0	0	7	6	2	Social Medium	0	0	7	11	14
11	France	0	0	2	1	0	Review	8	4	52	62	68
12	South Korea	0	0	4	1	2	Text	2	1	17	19	25
13	Canada	0	0	3	1	5	System	2	0	5	14	10
14	Turkey	0	0	2	1	2	COVID	0	0	0	3	1
15	Greece	0	0	4	2	1	Consumer	7	0	10	23	21
16	Malaysia	0	0	1	1	4	Crisis	0	0	8	7	11
17	Hong Kong	0	0	2	4	1	Sentiment Index	10	7	78	90	95
18	Poland	0	0	1	0	2	Shock	0	0	5	5	7
19	Portugal	0	0	1	0	1	Uncertainty	0	1	4	2	5
20	Netherlands	1	0	4	2	1	Algorithm	1	0	5	4	2

Note: All data were downloaded from Scopus as both CSV and plaintext, where all information was selected. This allows for the analysis of inward and outward citations, abstracts, and other bibliometric areas. Core bibliometric approaches involve surfacing the linkages between papers or articles. This data is subsequently separated and collated based on each Chartered Association of Business Schools (ABS) journal rankings list.

broader global interest in understanding the dynamics of these markets. However, an intriguing observation emerges regarding the under-representation of broad sectoral reviews and the evolution of sentiment indices. One plausible explanation is the rapid emergence of new methodologies and tools post-2010, making it challenging for traditional ranking systems to keep pace. These systems might possess a latency in recognizing the value of cutting-edge research areas or methodologies that deviate from the conventional paradigms. Moreover, the intrinsic dynamics of academic publishing, favoring incremental contributions to well-established fields, could inadvertently play a pivotal role compared to research areas on the periphery. As the significance of these overlooked areas becomes more pronounced, future evaluations are anticipated to reflect their rising importance in the academic landscape.

5 | DISCUSSION, IMPLICATIONS, AND DIRECTIONS FOR FUTURE RESEARCH

The meticulous scrutiny of sentiment-related studies within the economics and finance domains reveals crucial discussion points. Distinctly, there is an evident split in research trajectories: one

Table 6 Abstract and keyword occurrence, and geographical distribution of unique economics and finance publications (as ranked by ABDC).

Rank	Country	A*	A	B	C	Keyword	A*	A	B	C
1	United States	39	50	50	4	Market	44	114	154	42
2	China	17	62	33	4	Investor	41	97	134	29
3	United Kingdom	14	22	12	7	Return	30	85	98	19
4	India	0	6	27	10	Index	13	45	65	16
5	Germany	15	11	9	2	Risk	19	41	55	14
6	Australia	9	10	3	2	Stock Market	16	36	69	25
7	Spain	2	7	9	2	Volatility	15	42	65	16
8	Italy	5	6	14	3	Price	21	50	75	22
9	Russia	1	2	3	2	News	23	37	38	11
10	Taiwan	4	14	5	6	Social Medium	4	41	55	51
11	France	8	6	7	0	Review	44	114	154	42
12	South Korea	3	4	5	1	Text	19	41	43	15
13	Canada	4	5	5	0	System	6	28	18	13
14	Turkey	1	3	5	2	COVID	4	8	27	2
15	Greece	1	8	7	3	Consumer	8	23	12	6
16	Malaysia	0	3	4	4	Crisis	9	15	31	8
17	Hong Kong	1	7	5	0	Sentiment Index	83	171	212	67
18	Poland	1	1	4	3	Shock	10	15	21	2
19	Portugal	0	3	5	3	Uncertainty	11	11	30	3
20	Netherlands	2	3	3	0	Algorithm	6	9	7	5

Note: All data were downloaded from Scopus as both CSV and plaintext, where all information was selected. This allows for the analysis of inward and outward citations, abstracts and of a wide variety of other bibliometric areas. Core bibliometric approaches involve surfacing the linkages between papers or articles. This data is subsequently separated and collated based on the Australian Business Deans Council (ABDC) journal rankings list.

that delves deep into the underpinnings of sentiment theories and another that assesses their direct consequences on economic metrics. This split accentuates the imperative to revisit and reassess the interplay between theoretical constructs and their tangible ramifications in the market arena. Moreover, a pronounced focus on terms associated with market dynamics, risk, and investor sentiment unveils an emerging trend in research, converging on the intersection of sentiment and market equilibrium.

A striking observation is the limited representation of sentiment analysis in esteemed academic journals. This prompts an introspection into the criteria for academic endorsement and kindles a debate on the re-calibration of interdisciplinary research acknowledgment. The prevailing influence of nations, predominantly the United States, China, and the United Kingdom, in spearheading sentiment research propels discussions on the potential overshadowing of insights from emerging economies, emphasizing the urgency for diversified research synergies.

Furthermore, the dynamic capabilities of sentiment analysis, notably its agility in adapting to unpredictable global phenomena, necessitate a discussion on its potential elevation from an ancillary tool to a cornerstone analytical asset. Collectively, these insights underline the transformative trajectory of sentiment research, signaling the necessity for cohesive research methods and amplifying the importance of sentiment analysis in deciphering complex financial ecosystems.

Table 7 Abstract and keyword occurrence, and geographical distribution of unique economics and finance publications (as ranked by SJR).

Rank	Country	Q1	Q2	Q3	Q4	Keyword	Q1	Q2	Q3	Q4
1	United States	18	37	18	16	Market	13	54	63	27
2	China	2	8	14	3	Investor	7	46	35	26
3	United Kingdom	3	23	9	1	Return	4	45	34	15
4	India	0	1	1	1	Index	1	21	14	11
5	Germany	1	12	9	3	Risk	3	23	15	14
6	Australia	1	9	5	3	Stock Market	3	12	25	10
7	Spain	0	6	5	1	Volatility	3	15	20	9
8	Italy	0	1	6	4	Price	5	21	28	19
9	Russia	0	0	2	0	News	5	22	11	8
10	Taiwan	0	8	4	1	Social Medium	0	0	5	8
11	France	1	3	4	1	Review	13	54	63	27
12	South Korea	0	4	1	1	Text	3	19	13	9
13	Canada	0	5	4	0	System	2	5	13	5
14	Turkey	0	4	0	0	COVID	0	0	2	1
15	Greece	0	4	3	2	Consumer	6	8	12	6
16	Malaysia	0	0	2	1	Crisis	0	11	8	8
17	Hong Kong	0	3	1	0	Sentiment Index	19	96	88	48
18	Poland	0	1	0	0	Shock	1	7	8	5
19	Portugal	0	1	0	1	Uncertainty	1	8	4	3
20	Netherlands	2	3	2	1	Algorithm	0	4	4	0

Note: All data were downloaded from Scopus as both CSV and plaintext, where all information was selected. This allows for the analysis of inward and outward citations, abstracts and of a wide variety of other bibliometric areas. Core bibliometric approaches involve surfacing the linkages between papers or articles. This data is subsequently separated and collated based on the SCImago Journal Rank (SJR) journal rankings list.

This inquiry’s findings highlight the pivotal role of nations such as the United States, China, and the United Kingdom in guiding sentiment research narratives, suggesting that academic deliberations and resultant market strategies could lean heavily towards these regions. This concentration underscores the need for encompassing perspectives from a diverse array of economies to achieve a truly global financial understanding.

The relative absence of sentiment analysis in premier academic literature spurs a reevaluation of current benchmarks of scholarly excellence. The agile nature of sentiment analysis, its potential to address contemporary global challenges, and the recurrent emergence of certain key terminologies accentuate its foundational role in modern economics. This suggests that there is a compelling need for academic paradigms to evolve and for the broader financial community to leverage sentiment analysis more proactively. In sum, this investigation underscores the importance of collaborative academic endeavors, redefining scholarly benchmarks, and capitalizing on sentiment analysis to craft foresighted financial strategies.

The rigorous assessment of sentiment-related studies within the spheres of economics and finance unveils salient avenues warranting future exploration. One evident trajectory emerges from the observed bifurcation in research themes: the foundational theories of sentiment and their tangible repercussions on market dynamics. This dichotomy underscores a pressing need to

bridge these seemingly disparate realms, fostering a synthesis that could lead to a more holistic understanding of sentiment's role in shaping market outcomes.

Moreover, the conspicuous under-representation of sentiment analysis within esteemed academic publications raises pivotal questions about prevailing academic validation benchmarks. Future research might thus consider critically examining these benchmarks, discerning whether they fully encapsulate the dynamism and adaptability of interdisciplinary approaches such as sentiment analysis. There is a palpable urgency for investigations that determine the criteria of academic endorsement and explore avenues to enhance the prominence of sentiment research within these metrics.

Additionally, the demonstrated adaptability of sentiment analysis in responding to unforeseen global events propels discussions on its broader applications. As the world faces increasing uncertainties, from economic downturns to global health crises, there is a compelling need to explore how sentiment analysis can be utilized beyond its current capacities. Future research might focus on developing more sophisticated analytical tools that harness the predictive power of sentiment analysis, examining its efficacy in anticipating market shifts and proactively informing policy decisions.

In sum, the findings from this study illuminate a tapestry of research trajectories. They call for a more integrative approach in sentiment research, a reassessment of academic validation paradigms, the necessity for diverse collaborative undertakings, and the exploration of sentiment analysis as a vanguard tool in navigating an increasingly unpredictable financial landscape. These avenues, when pursued, hold the potential to reshape the contours of future economic and financial research.

6 | CONCLUDING COMMENTS

Real-time stock analysis and price prediction remain a big challenge, as an algorithm might work in controlled environments but not in practice due to price variations, uneventful news and noise (Shah et al., 2019; Xu & Cohen, 2018). The existing online predictive models need improvements to be updated with data in real-time. Field-specific dictionaries need to be updated for the lexicon-based methods. Another issue to be addressed is the sudden rare changes such as the uncertainty inherent within the COVID pandemic (Chou et al., 2021). Furthermore, the effect of the sentiment expressed in different languages can be considered by using multiple languages in sentiment analysis. Finally, the ML methods need to be applied to the cryptocurrency market, which is more volatile compared to the conventional stock market. The examination of sentiment-related research in economics and finance domains has unearthed several pivotal findings. To begin with, there's an undeniable centrality of specific nations, particularly the United States, China, and the United Kingdom, in spearheading the academic discourse in this field. Their dominant position in the global landscape can be attributed to a confluence of factors: robust academic infrastructure, significant research funding, and a history of pioneering work in finance and economics.

While the scope of MSA research has historically encompassed traditional financial markets, recent years have witnessed a marked shift towards analyzing emerging financial ecosystems, particularly cryptocurrencies. These volatile and speculative digital assets present unique challenges and opportunities for sentiment analysis. Unlike traditional markets, cryptocurrencies are highly influenced by social media dynamics and investor sentiments, making them a fertile ground for advanced MSA techniques. This trend reflects not only the growing popularity of digital currencies but also highlights the adaptability of MSA methodologies to novel and rapidly evolving financial

environments. The exploration into cryptocurrency markets has broadened the application scope of MSA and sparked interest in understanding how market sentiment interacts with new forms of financial innovation and investment behavior.

Another consequential observation is the crystallisation of certain keywords - “market”, “investor”, “return”, “index”, and “risk”. Their recurrent appearance denotes a thematic underpinning of current sentiment-related research towards understanding market dynamics, investor behavior, and risk assessment. It is also evident that sentiment analysis is increasingly recognized as a standalone research area and a tool to gauge market sentiment, understand price volatility and evaluate liquidity effects. Furthermore, the emergence of the term “COVID” in recent literature underscores the flexibility and adaptability of sentiment analysis. It is a testament to the field’s capability to integrate contemporary events and their far-reaching implications on global economies. This responsiveness ensures its relevance, especially when unforeseen global events, such as pandemics, challenge existing economic models and paradigms. However, while these findings offer a comprehensive understanding of the current landscape, they also hint at potential gaps. The conspicuous under-representation of research in high-quality academic outputs, as highlighted by rankings such as ABDC, SJR, and ABS, indicates a possible misalignment between perceived academic rigor and the groundbreaking nature of sentiment analysis research.

In essence, this research not only maps the current terrain of sentiment-related studies in economics and finance but also underscores the dynamic interplay between evolving global events, market dynamics, and academic research, underpinning the enduring relevance of sentiment analysis, its adaptability, and its potential to redefine the contours of future economic and financial research.

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ENDNOTES

¹Tech Shares Rise With Bond Yields on Biden’s Plan: Markets Wrap, Bloomberg (accessed online on February 2, 2024), available <https://www.bloomberg.com/news/articles/2021-03-30/asia-set-for-mixed-open-oil-drops-dollar-gains-markets-wrap?srnd=premium-europe> here and Biden Infrastructure Plan Targets Electric Cars, Clean Power (accessed online on February 2, 2024), available <https://www.bloomberg.com/news/articles/2021-03-31/biden-infrastructure-plan-would-boost-electric-cars-clean-power?leadSource=uverify%20wallhere>.

²Bitcoin price back above \$40,000 after Elon Musk comments”, The Guardian (accessed online on February 2, 2024), available <https://www.theguardian.com/technology/2021/jun/14/tesla-will-resume-accepting-bitcoin-when-it-uses-more-clean-energy-says-muskhere>.

³Many works are separated using corporate market capitalization, which is a measure of the value of a company in the market, determined by multiplying the market share price by the total number of shares. It shows how much a company is worth in the market. This is related to the company’s assets, technologies, patents, workforce, and other factors. Companies are typically divided into three bands, based on market capitalisation: large-cap (\$10 billion or more), mid-cap (\$2 billion to \$10 billion), and small-cap (\$300 million to \$2 billion).

⁴From the author’s knowledge of the area, we are confident that no significant academic source of papers was omitted.

⁵We are using the terms “bibliometrics”, “informetrics”, and “scientometrics” interchangeably; see Abramo and Oxley (2021).

⁶Waltman (2016) contains an extensive discussion of self-citation and its effects on scientometric measures and analyses, concluding that in large-scale analyses, self-citation does not overly bias or distort findings. Nor is there any clear finding that for authors, as opposed to countries or institutions, self-citation should pose a problem for scientometric analysis overall.

- ⁷The last two exclusions were required due to large numbers of papers being returned, which were opinions or reportage from *The Economist Newspaper*, classified as articles by Scopus.
- ⁸These networks can be constructed based on citation, bibliographic coupling, co-citation, or co-authorship relations. VOSviewer also offers a text mining functionality that can be used to construct and visualize co-occurrence networks of important terms extracted from a body of scientific literature.
- ⁹For those seeking more information on using such techniques in business and economics, see Rousseau and Rousseau (2021).
- ¹⁰Thus, linkages are stronger when, say, two authors collaborate on 10 papers than would be the case where another two authors collaborate on five. These linkages lend themselves nicely to graphical presentation, being, in essence, network models. Graphic models rely on nodes and edges (Kosnik, 2018), where the nodes are determined by the individual units of analysis (authors, countries, etc.) and the edges by the linkages between them. In all cases, we apply fractional counting, whereby authorship or nationality, among other characteristics, are scaled to the number of occurrences. Therefore, an author appearing in a paper with five others has their linkages weighted $\frac{1}{6}$.
- ¹¹For analysis of centrality measures and checking of the consistency of the graphs generated from Vosviewer
- ¹²For preliminary data analysis and measures of author and country dominance
- ¹³The Journal of Economic Surveys special issue on Informetrics, which raises a number of such points, see in particular D'Ippoliti (2021).
- ¹⁴In furthering our exploration of research grading systems, it is essential to acknowledge the diverse spectrum of evaluation methodologies applied across different academic and professional spheres. Systems, for example, the Australian Business Deans Council (ABDC), the Academic Journal Guide (AJG) by the Chartered Association of Business Schools (ABS), and SCImago Journal Rank (SJR) represent prevalent models. Each system adopts unique assessment criteria, ranging from citation impact and journal prestige to practical applicability and methodological rigor. While ABDC focuses predominantly on the relevance to business and management domains, AJG emphasizes methodological precision and theoretical contribution. On the other hand, SJR offers a more global perspective, incorporating international visibility and interdisciplinary reach. These variations in grading criteria highlight the multifaceted nature of academic excellence, urging scholars and practitioners alike to consider a broader range of factors when evaluating the impact and significance of research in economics and finance.

ACKNOWLEDGMENTS

Open access publishing facilitated by The University of Waikato, as part of the Wiley - The University of Waikato agreement via the Council of Australian University Librarians.

CONFLICT OF INTEREST STATEMENT

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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How to cite this article: Heydarian, P., Bifet, A., & Corbet, S. (2025). Understanding market sentiment analysis: A survey. *Journal of Economic Surveys*, 39, 1125–1147. <https://doi.org/10.1111/joes.12645>